

一般约束优化问题

$$(P) \quad \begin{aligned} \min_x & f(x) \\ \text{s.t.} & c_{ii}(x) = 0 \quad i \in \mathcal{I} \\ & c_{2j}(x) \leq 0 \quad j \in \mathcal{J} \end{aligned}$$

罚方法：二次罚方法是不可行方法，迭代过程中的可行性不能相互满足

$$(Pen) \quad \min_x f(x) + \frac{\sigma}{2} (\|c_1(x)\|_2^2 + \|\max(c_2(x), 0)\|_2^2)$$

Thm: x^{k+1} 是 $P_{\sigma^k}(x)$ 的全局极小解 $\sigma^k \rightarrow +\infty, k \rightarrow +\infty$
 $\Rightarrow \{x^k\}$ 的极限点 x^* 是原问题的全局极小解

注：“ x^{k+1} 是 $P_{\sigma^k}(x)$ 的全局极小解”这一条件很难满足

→ 问题的终止准则

Thm: 设 $f(x), c(x)$ 可微, 若 x^{k+1} 满足 $\|\nabla_x P_{\sigma^k}(x^{k+1})\| \leq \varepsilon_k$, 且 $\varepsilon_k \rightarrow 0, \sigma^k \rightarrow +\infty$. 对 $\{x^k\}$ 的任何极限点 x^* , 满足 LICQ, 则 x^* 是原问题的 KKT 点, 且

$$\lim_{k \rightarrow +\infty} (-\sigma^k c_i(x^{k+1})) = \lambda_i^*$$

$$\begin{aligned} \min_x & f(x) \\ \text{s.t.} & c_i(x) = 0 \end{aligned}$$

$$P_{\sigma}(x) = f(x) + \frac{\sigma}{2} \|c(x)\|_2^2 \quad \text{其中 } \lambda_i^* \text{ 是 } c_i(x^*) = 0 \text{ 的 Lagrangian 乘子.}$$

Proof: 由终止准则, 可得

$$\|a\| - \|b\| \leq \|a + b\|$$

$$\|\nabla_x P_{\sigma^k}(x^{k+1})\| = \|\nabla f(x^{k+1}) + \sigma^k [\nabla c(x^{k+1})]^T c(x^{k+1})\| \leq \varepsilon_k$$

$$\Rightarrow \sigma^k \|[\nabla c(x^{k+1})]^T c(x^{k+1})\| - \|\nabla f(x^{k+1})\| \leq \varepsilon_k$$

$$\Rightarrow \|[\nabla c(x^{k+1})]^T c(x^{k+1})\| \leq \frac{1}{\sigma^k} (\varepsilon_k + \|\nabla f(x^{k+1})\|)$$

$\therefore k \rightarrow +\infty, \sigma^k \rightarrow +\infty$, 可得

$$\| [\nabla c(x^*)]^T c(x^*) \| = 0$$

由于在 x^* 处满足 LICQ, 可得 $c(x^*) = 0$, 故 x^* 为可行点.

构造 Lagrangian 乘子: $\lambda_i^k = -\sigma^k c_i(x^{k+1})$, 且

$$\nabla_x P_{\sigma^k}(x^{k+1}) = \nabla f(x^{k+1}) - [\nabla c(x^{k+1})]^T \lambda^k$$

$$\Rightarrow [\nabla c(x^{k+1})]^T \lambda^k = \nabla f(x^{k+1}) - \nabla_x P_{\sigma^k}(x^{k+1})$$

由于 LICQ, $\nabla c(x^*)$ 行满秩, 当 k 足够大时, $\nabla c(x^{k+1})$ 也是行满秩

$$\lambda^k = [\nabla c(x^{k+1}) \nabla c(x^{k+1})^T]^{-1} \nabla c(x^{k+1}) [\nabla f(x^{k+1}) - \nabla_x P_{\sigma^k}(x^{k+1})]$$

由于 $\|\nabla_x P_{\sigma^k}(x^{k+1})\| \leq \sigma^k \rightarrow 0$, 有

$$\left\{ \begin{aligned} \lambda^* &= \lim_{k \rightarrow +\infty} \lambda^k = [\nabla c(x^*) \nabla c(x^*)^T]^{-1} \nabla c(x^*) \nabla f(x^*) \\ \nabla_x P_{\sigma^*}(x^*) &= \nabla f(x^*) - \nabla c(x^*)^T \lambda^* = 0 \end{aligned} \right.$$

可得 KKT 条件成立, λ^* 为对应的 Lagrangian 乘子, 且

$$\lim_{k \rightarrow +\infty} [-\sigma^k c_i(x^{k+1})] = \lambda_i^*$$

注: 要求 $\sigma^k \rightarrow +\infty$ 时, 才能满足可行性.

$$c_i(x^{k+1}) \approx -\frac{\lambda_i^*}{\sigma^k} \quad \sigma^k \rightarrow +\infty, \text{ 有 } c_i(x^{k+1}) \rightarrow 0$$

推广 Lagrangian 方法 (ALM, Augmented Lagrangian Method, 乘子罚函数)

$$\begin{aligned} \min_x & f(x) \\ \text{s.t. } & c_i(x) = 0 \end{aligned}$$

$$\text{二次罚: } f(x) + \frac{\sigma}{2} \|c(x)\|^2$$

$$\text{ALM: } f(x) + \lambda^T c(x) + \frac{\sigma}{2} \|c(x)\|^2$$

$$f(x) - \lambda^T c(x) + \frac{\sigma}{2} \|c(x)\|^2 \leftarrow \text{也可以这样写}$$

Thm: 设 $x^* = \arg \min \{f(x) \mid c(x) = 0\}$, λ^* 为最优乘子, 且在 x^* 处满足 LICQ, 则

$$L_{\sigma}(x; \lambda) = f(x) + \lambda^T c(x) + \frac{\sigma}{2} \|c(x)\|^2$$

存在 $\sigma^* > 0$, 使得任意 $\sigma > \sigma^*$, x^* 为 $L_{\sigma}(x; \lambda)$ 严格最小值点.

注: ALM 无需 σ 趋于无穷, 也可得到原问题的最优解.

缺点: 无法保证 λ 的取值, 需要每步迭代计算 λ .

变量个数: $n \rightarrow n+m$.

算法: ALM 方法.

1. 给定 x^0 , $\lambda^0 = 0$, $\sigma^0 > 0$, 取 $\varepsilon > 0$, $p > 1$.
2. for $k=1, 2, \dots$
3. 求 $x^{k+1} = \arg \min_x L_{\sigma^k}(x; \lambda^k)$
4. 更新乘子 $\lambda^{k+1} = \lambda^k + \sigma^k c(x^{k+1})$
5. 更新罚参数 $\sigma^{k+1} = p \sigma^k$
6. If $\|c(x^k)\| \leq \varepsilon$ then break. End If.
7. End for

如何更新乘子?

在第 k 步迭代:

$$\nabla_x L_{\sigma^k}(x^{k+1}; \lambda^k) = \nabla f(x^{k+1}) + [\lambda^k + \sigma^k c(x^{k+1})]^T \nabla c(x^{k+1}) = 0$$

最优解 x^* 处:

$$\nabla f(x^*) + [\lambda^*]^T \nabla c(x^*) = 0$$

← ALM 希望收敛

因此, 对于充分大的 k ,

$$\lambda^* \approx \lambda^k + \sigma^k c(x^{k+1})$$

$$\Rightarrow c(x^{k+1}) \approx \frac{1}{\sigma^k} (\lambda^* - \lambda^k) \leftarrow \text{当 } \lambda^* \text{ 和 } \lambda^k \text{ 足够接近}$$

可作为乘子的更新公式

算法2: Adaptive ALM 方法

迭代远离可行域
→
可行域
→
加入罚项
→
迭代靠近可行域
→
更新乘子

1. 给定 x^0 , $\lambda^0 = 0$, $\sigma^0 > 0$, 取 $\varepsilon > 0$, $\rho > 1$.

2. 求 $x^{k+1} = \arg \min_x \mathcal{L}_{\sigma^k}(x; \lambda^k)$

3. If $\|c(x^{k+1})\| < \varepsilon$, then break End If

4. If $\|c(x^{k+1})\|_{\infty} \geq \|c(x^0)\|_{\infty}$ then

5. $\sigma^{k+1} = \rho \sigma^k$, $\lambda^{k+1} = \lambda^k$, $k = k+1$, 转 step 2

6. Else If $\sigma^k > \sigma^{k-1}$ or $\|c(x^{k+1})\|_{\infty} \leq \frac{1}{\rho} \|c(x^0)\|_{\infty}$

7. $\sigma^{k+1} = \sigma^k$, $\lambda^{k+1} = \lambda^k + \sigma^k c(x^{k+1})$, $k = k+1$. 转 step 2

8. End if

不等式约束 ALM,

$$\begin{aligned} \min_x & f(x) \\ \text{s.t.} & c(x) \leq 0 \end{aligned}$$

$$\mathcal{L}_{\sigma^k}(x, y; \lambda) = f(x) + \lambda^T (y^2 + c(x)) + \frac{\sigma}{2} \|y^2 + c(x)\|_2^2$$

$$\text{问题} \quad (x^{k+1}, y^{k+1}) = \arg \min_{(x, y)} \mathcal{L}_{\sigma^k}(x, y; \lambda^k)$$

$$\Rightarrow \min_x \min_y \mathcal{L}_{\sigma^k}(x, y; \lambda^k)$$

$$\Rightarrow \min_x \min_{s \geq 0} \mathcal{L}_{\sigma^k}(x, y; \lambda^k) \quad s = y^2$$

$$\min_x \min_{s \geq 0} f(x) + \lambda^T (s + c(x)) + \frac{\sigma}{2} \|s + c(x)\|_2^2$$

$$\lambda + \sigma(s + c(x)) = 0 \quad s \geq 0$$

$$s = \max\left(-c(x) - \frac{\lambda}{\sigma}, 0\right)$$

$$\Rightarrow \underline{c(x) + s} = \max\left(-\frac{\lambda}{\sigma}, c(x)\right)$$

$$\lambda^T (c(x) + s) + \frac{\sigma}{2} \|c(x) + s\|_2^2$$

$$= \begin{cases} -\frac{1}{2\sigma} \|\lambda\|^2 & \text{if } -\frac{\lambda}{\sigma} > c(x) \\ \lambda^T c(x) + \frac{\sigma}{2} \|c(x)\|_2^2 & \text{otherwise} \end{cases}$$

$$\Rightarrow P_i(x; \lambda) = \begin{cases} -\frac{1}{2\sigma} \lambda_i^2 & \text{if } \frac{\lambda_i}{\sigma} > c_i(x) \\ \lambda_i c_i(x) + \frac{\sigma}{2} c_i(x)^2 & \text{otherwise} \end{cases}$$

$$\Rightarrow L_{\sigma}(x; \lambda) = f(x) + \sum P_i(x; \lambda)$$